

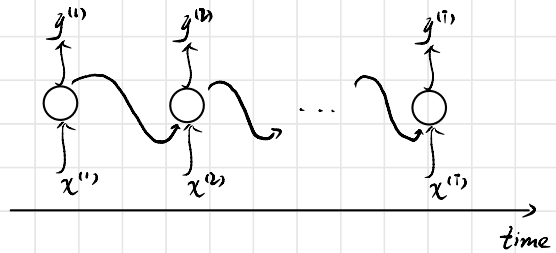
Before Transformer, attention mechanism exists to solve the short term memory issues with RNN.

Simple RNN

One neuron:



unroll through time $\rightarrow$

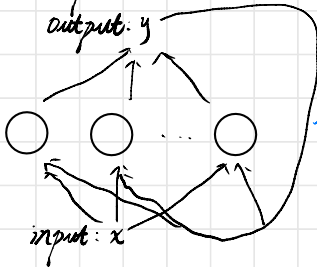


formulation:

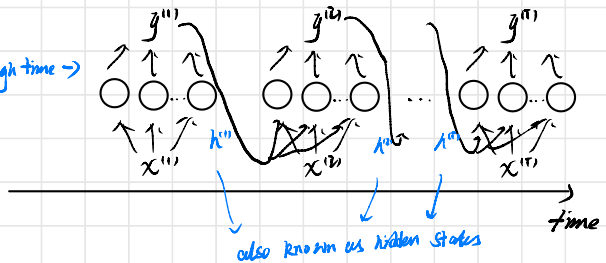
$$y^{(t)} = \sigma \left(v_h y^{(t-1)} + v_x x^{(t)} + b \right)$$

Annotations:
 σ : activation function
 v_h : a scalar
 v_x : a scalar
 $x^{(t)}$: a vector of length d (the dimension of inputs)

Multiple neurons



unroll through time $\rightarrow$



formulation:

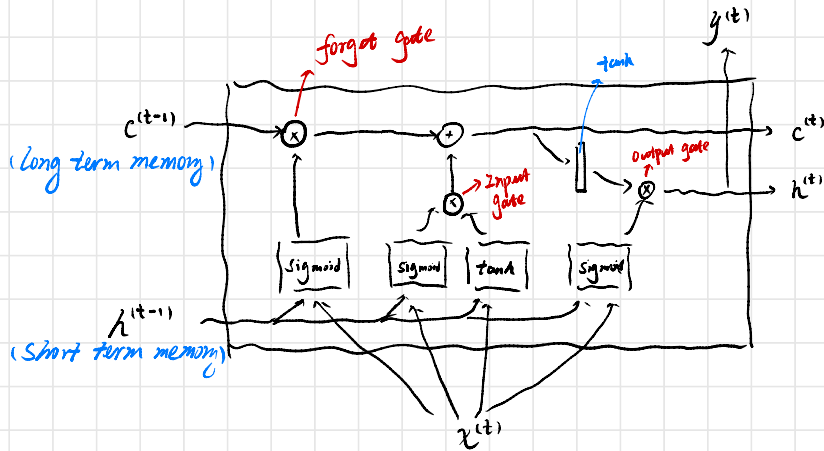
$$y^{(t)} = \sigma \left(W_h^T y^{(t-1)} + W_x^T x^{(t)} + b \right)$$

Apparently, we can stack many layers of RNN to make our model more representative.

problem with simple RNN: Short-term memory. $y^{(t)}$ contains less and less information about $x^{(1)}$ when t increases.

LSTM (long short term memory) cells solve the short term memory problem partly by introducing **gates** to divide memory into short memory and long memory.

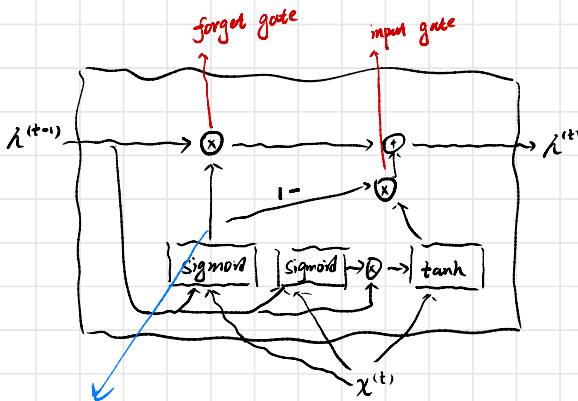
LSTM cell



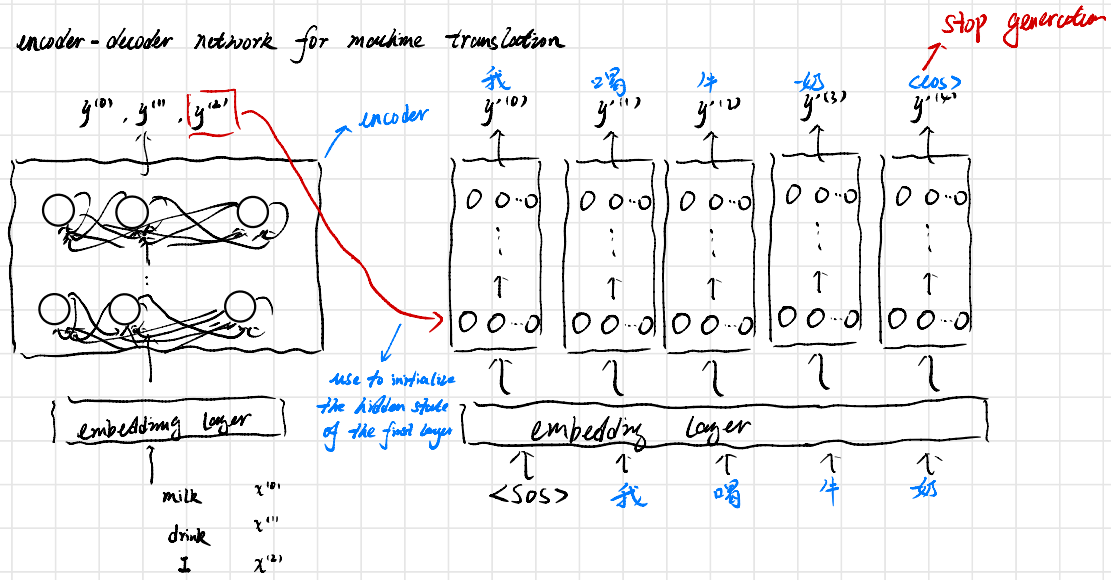
pros of LSTM: it decides what information to forget and keep so in theory, important information can be kept.

Cons of LSTM: it is very expensive to compute

A simplified version of LSTM: GRU (gated recurrent units)

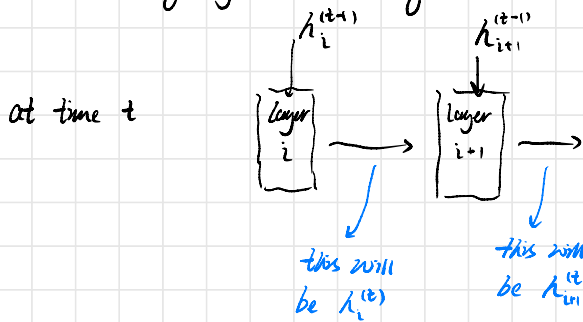


An encoder-decoder network for machine translation



notice this is reversed.
because I is normally the
first word to translate.

Decoder during inference: autoregressive.



each layer deals with
one time step and moves
immediately to next layer

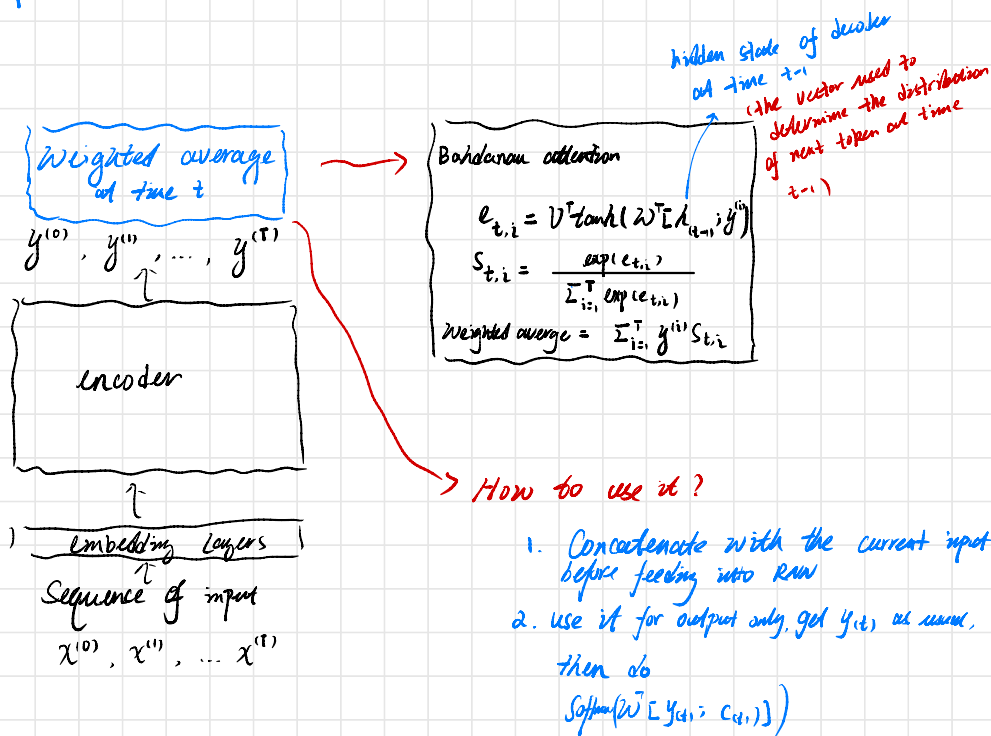
Decoder during training: behave exactly like encoder by applying teacher forcing.

each layer process all time steps and then
move to the next layer.

problems with standard encoder-decoder: information bottleneck
need to encode whole sentence into a fixed length vector. when the
sequence become longer, earlier information will be deleted.

for predicting $y^{(n)}$, we
assume the model predicts
 $y^{(1)}, \dots, y^{(n-1)}$ correctly

Bahdanau attention: In standard encoder-decoder, only the last embedding of the encoder is used for decoding, because it is the only embedding has all information of the input. In Bahdanau attention, at each time step, all embeddings of encoder will be weighted average. This new embedding determines which word should the decoder to focus at this time step.



Why is attention mechanism revolutionary?

Break the information Bottleneck.

With attention, the whole sentence (3 words long or 100 words long) are compressed into a fixed length vector. With attention, the information can be represented into number of words $\times$ vector length. So it scale dynamically.